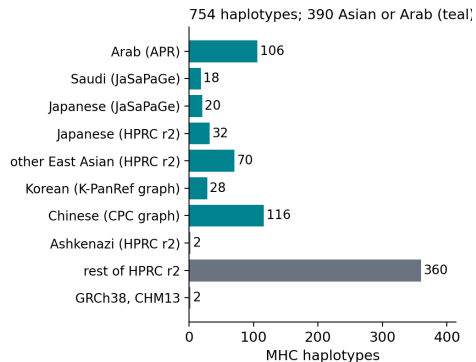
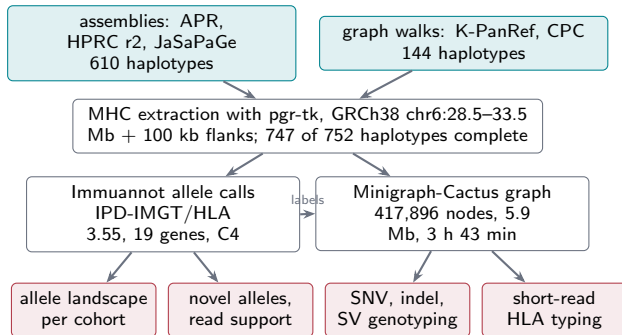
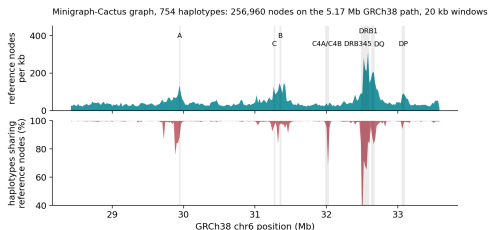


An Asian-enriched MHC pangenome: 754 haplotypes from five projects, one graph, three uses

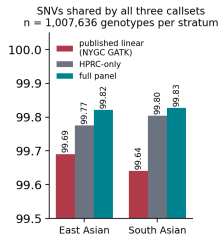
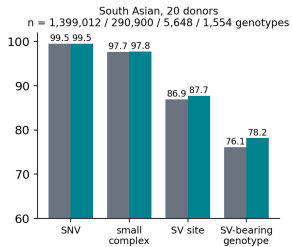
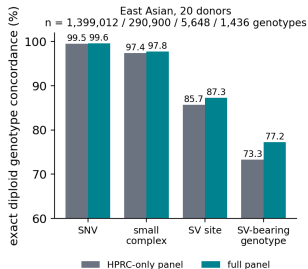


- ▶ Aim 1: one MHC graph that adds Arab, Saudi, Japanese, Korean and Chinese haplotypes to HPRC release 2, each labelled with full-resolution HLA alleles
- ▶ Aim 2: do the added haplotypes improve genotypes from 30x Illumina reads? Control: an HPRC-only panel run through the same method
- ▶ Aim 3: an HLA typer that names alleles and returns the full gene sequence, benchmarked against T1K and SpecHLA on 1000 Genomes donors whose own assemblies are the truth

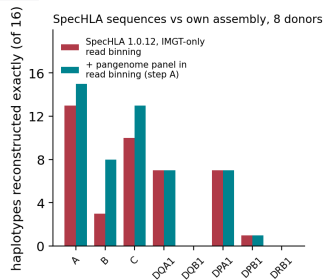
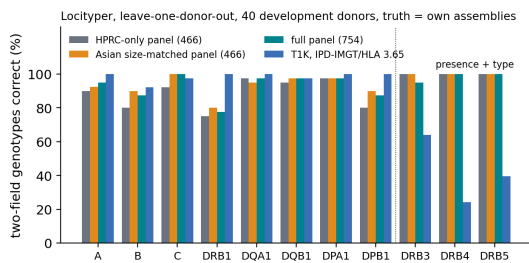
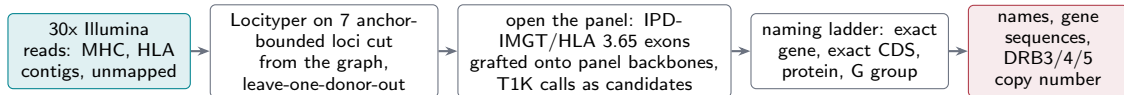
The added haplotypes improve every variant class in held-out donors, most for SVs



- Left: along the GRCh38 path the graph is near-linear except at HLA-A, C and B, C4, the DR-DQ block and DP; only in the DRB3/4/5 block do fewer than 60% of haplotypes share the reference nodes. vg giraffe maps 99.5% of one sample's MHC read pairs in 26 s
- Below: PanGenie 4.2.1, identical MHC reads of 40 held-out 1000 Genomes donors, truth = the donor's own assembly in the graph, test families excluded from both panels, same sites. Full vs HPRC-only: +0.1 (SNV) to +3.9 points (SV-bearing genotypes, East Asian; 95% CI +2.4 to +5.3); on SNVs shared with the published NYGC GATK callset, +0.13 and +0.19 points over linear



Short-read HLA typing: panel wins on DRB3/4/5 and sequences, T1K on two-field names



- ▶ Left, 40 development donors: classical 8 genes two-field, full panel 92.5%, Asian size-matched 92.8%, HPRC-only 88.4%, T1K 98.4%; DRB3/4/5 with copy number, panel 111/113, T1K 48/110. Right: the panel in SpecHLA's read binning raises exact gene reconstructions from 41 to 51 of 128 haplotypes (HLA-B 3 to 8); names unchanged
- ▶ Remaining errors: DRB1 picks and alleles absent from the panel; grafted IPD exons recover 4 so far. The locked test (228 leave-one-out donors, 946 Gourraud-typed donors) has not run yet

Truth = the donor's own assembly haplotypes, IPD-IMGT/HLA 3.65.0 labels; no-calls count as errors; T1K on the same reads and database release. Code: [hla-typer/](#), [hla-spechla-pg/](#)